Durability

- **Durability** is the wood's ability to resist decay from exposure to the elements and attack from insects.
- Hardwoods such as teak are the most durable woods and can last for many years when exposed to the elements.
- Some types of bamboo have a durability of only a few months.

Durability: resistance to wear, long lasting.

Now test yourself

TESTED

1 Why is plywood a much stronger manufactured board than MDF? (2)
2 Name **one** advantage and **one** disadvantage of knots in pine. (4)
3 Explain **one** major difference between hardwood and softwood trees. (2)

1.13 Contexts which inform design outcomes

1.13.1 Discriminating between materials, components and manufacturing processes

REVISED

Properties may change depending on things such as temperature or the shape and direction of the material. Other properties will remain the same regardless of the environment or conditions.

Mechanical properties include:

- compressive strength – resistance to a pushing force
- tensile strength – resistance to a pulling force
- durability – resistance to wear, long lasting
- hardness – resistance to abrasion
- stiffness – resistance to bending and flexing
- elasticity – ability to stretch and return to shape
- impact resistance – resistance to knocks and impacts
- brittleness – hard but easily broken or shattered
- water resistance – resistance to water or moisture
- absorbency – ability to soak up moisture.

Manufacturing properties include:

- machinability – ability to be easily milled and drilled
- workability – ability to be placed and compacted
- ductility – ability to stretch, not break under stress or impact
- malleability – ability to be hammered, pressed or rolled without breaking.

Other properties include:

- acoustic – acoustic absorption
- chemical – corrosion resistance
- electrical – conductivity, resistance and so on
- thermal – conductivity, flash point and so on
- magnetic – attractive to magnets.

Typical mistake

Malleability and ductility are similar properties and often confused. Ductility relates to the ability of a material to stretch without damage. Malleability relates to the ability of a material to deform under compression.

Exam tip

By learning the properties of materials, you can understand why they have been used to make certain products. This will help you to answer questions in the exam about individual products and their fitness for purpose.

Advantages and disadvantages

- The choice of materials impacts the success of the final product.
- The advantages of a particular material may make it an obvious choice for a product, but the designer must also consider disadvantages, for example a thermosetting polymer may have the best properties for a product but it is not recyclable. Therefore, the designer may choose a thermoforming polymer with properties as close to the thermosetting one as possible.

Justification of the choice

- The designer must make informed choices based on these characteristics and other factors, such as:
  - cost – is there another material available that may cost less?
  - availability – is the material in good supply?
  - sustainability – is the material sustainable?
  - environmental impact – can the material be recycled?
  - social, cultural and moral considerations – will the use of this material have a negative impact?
- When designing an electronic system, the designer must consider the range of components available and decide on the most suitable type for the application.

Now test yourself

TESTED

1 State **four** important mechanical properties of materials. (4)
2 State **two** factors that can influence and affect the properties of materials. (2)

1.14 Environmental, social and economic challenges

1.14.1 Respect for different social, ethnic and economic groups

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Positive and negative impacts of products

- Products can have a negative impact on society that the designer may not have foreseen.
- For example, leaflets and flyers can be used to communicate useful information to people about services in the area, however this can lead to unwanted junk mail being delivered, urging people to buy things they don't want.
- While it is not always easy to predict the effects of a product, designers must try to consider the possible negative implications and decide whether the product is worthwhile.

- physical size and weight of the product being manufactured
- production capacity of the manufacturing plant.

1.14.8 Life-cycle analysis

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- The life cycle of a product is the stages a product goes through from beginning to end.
- A life-cycle analysis (LCA) assesses the environmental impacts of a product at the following stages:
  - extraction of the raw materials
  - processing of the materials into useable form
  - manufacture of the product
  - distribution and transportation to retailers
  - use by the customer
  - repair and maintenance of the product
  - disposal or recycling of the product.

Exam tip

Practise drawing LCAs for a range of products.

Now test yourself

TESTED

1 State **three** factors that may be considered when designing recyclable products. (3)

2 Name **three** sustainable materials and explain why they are sustainable. (6)

3 Describe **one** factor that can influence human capability. (2)

1.15 The work of past and present professionals and companies

1.15.1 Analysing a product to specification criteria

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Table 1.13 Specification criteria

Criteria	Explanation
Form	The shape and appearance of a product
Function	The purpose of a product; what it will be used for
Client/user requirements	The needs, wants and values that are important to the people who will use it
Performance requirements	Factors that will help it to do its job; how it will perform when in use
Materials and components/ systems	What the product is made from and why; the properties the materials have that support function; how the product operates (system)
Scale of production and cost	The number of products to be manufactured will affect the scale of production and costs
Sustainability	The environmental impact from origin to disposal
Aesthetics	The visual appearance of the product, for example colour, texture
Marketability	How likely the product will sell; features that will help product sales
Innovation	Does it have any radically new features or include the latest technology?

1.16 Design strategies

1.16.1 Use of different design strategies

REVISED

- Innovative design is about exploring new ways of doing things or trying to approach problems from different angles or perspectives.
- **Design fixation** is when designers stick to one idea instead of trying to explore new design avenues. Designers can use different strategies to help them avoid design fixation.

Design fixation: when designers stick to one idea instead of exploring new design avenues.

Collaboration: working with others on a task.

Collaboration

- A design business will usually employ a number of designers. By discussing, sharing and working with each other they can get a different perspective on a problem.
- **Collaboration** allows designers to bounce ideas back and forth between each other. It can open new avenues of investigation and designs that one designer would not have thought of on their own.

Exam tip

Practise using different design strategies in class. This will help you understand the benefits of each.

User-centred design

- **User-centred design** is where the needs, wants and requirements of the **user** are looked at and checked at every stage.
- Putting the user at the centre of the design process means less refinement and modifications are likely to be needed later.
- The process has four main stages:
  1. **Context of use**: identify the users of the product, what they will use it for, and under what conditions it will be used.
  2. **Requirements**: identify any user goals that must be met.
  3. **Create design solutions**: this may be done in stages, building from a rough concept to a complete design.
  4. **Evaluate designs**: evaluation through usability testing with actual users.

User-centred design: the needs and wants of the user are considered at every stage.

User: the person or group of people a product is designed for.

Systems thinking

Systems thinking is a top-down method of designing, analysing or describing how the parts of a system work together to achieve a particular function. These parts are often broken down into input, process and output subsystems. In design and technology, this strategy is applied mainly to electronic and mechanical systems but can be used to help understand other, wider human-made and natural settings, for example natural ecosystems.

Now test yourself

TESTED

1 Explain **one** benefit of collaborative design work. (2)

2 Describe how user-centred design can save time when designing products. (2)

1.17 Developing, communicating, recording and justifying design ideas

1.17.1 Using a range of communication techniques and media

REVISED

Sketching

- **Freehand sketching** is a method of presenting design ideas on paper without the use of drawing aids or formal drawing conventions. It is often used to produce early ideas for products or as a tool for ideas generation. Sketching can be done in either 2D or 3D.
- Annotations are used to give further details and information about the ideas that have been sketched. This could cover areas such as:
  - the inspiration behind each design
  - the good and bad points
  - manufacturing methods that could be used to make each idea
  - how well each design meets the requirements of the **end user** and/or design specification.

Freehand sketching: method of presenting design ideas on paper without the use of drawing aids or formal drawing conventions.

End user: the person who will purchase or use the finished product.

Cut and paste techniques

Cut and paste techniques are used to make collages. Pictures and images are cut from different sources, such as newspapers, magazines and photographs, and glued on to paper to create an art form.

Digital photography/media

Digital technologies are increasingly being used in the design of products. For example, taking photos of iterative models to record each stage of development, or videos of testing to demonstrate how a design works. These can be shared online with the end user or other potential customers.

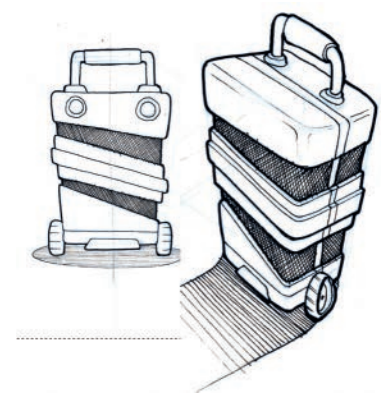


Figure 1.33 3D sketching

3D models

The use of **3D models** allows design ideas to visualised, handled and tested. Models allow the designer to spot any errors or areas for improvement before money is spent on more expensive materials. They can also be shared with the end user to gain their feedback before the final product is manufactured.

3D model: 3D version of a design idea made using cheap materials, for example card and Styrofoam™, used to check how it would look and function.

Isometric projection: method of drawing objects in 3D by drawing the sides at a 30-degree angle.

Isometric and oblique projections

- These are methods of drawing objects in 3D.
- In **isometric projections** the sides are drawn at a 30-degree angle.
- Oblique projections are simpler to produce than isometric, but not as realistic. The front side of the object is drawn first and lines are then drawn backwards at 45 degrees. The object is inclined either to the left or to the right.

Assembly drawings

Assembly drawings show how all the different parts of a product fit together. Each part is identified by a number. A bill of materials is included that lists each part needed and the quantities required.

Assembly drawing: drawing that shows how all the different parts of a product fit together, with each part identified by a number.

Schematic diagram: uses symbols to show a circuit in terms of its individual components and how they are connected.

System and schematic diagrams

- System, or block, diagrams show a system in terms of its input, processes and output subsystems. This presents a top-down overview of the system.
- **Schematic diagrams** show a circuit in terms of the individual components used. These are represented using symbols; the connections between them are shown with straight lines.

Computer-aided design

CAD software can be used to produce both drawings and virtual models of designs. Using CAD has many advantages, such as:

- Accuracy of drawing and modelling is very high.
- Changes can be made quickly and easily.
- No materials or physical resources are needed, reducing costs.
- Models can be simulated on screen.
- 2D drawings can be produced directly from the 3D model, saving time.
- CAD files can be outputted directly to CAM equipment for manufacture.

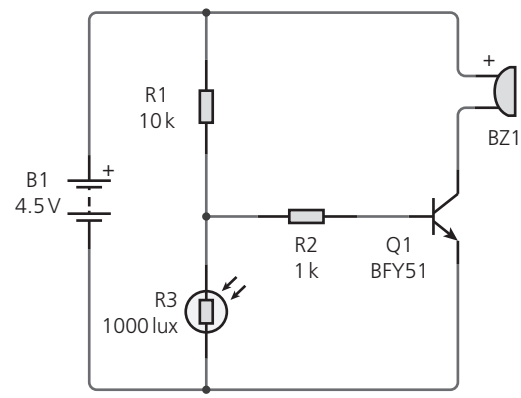


Figure 1.37 An electronic transistor circuit schematic

Exam tip

Make sure you know the purpose and correct use of each of the various ways of developing and communicating design ideas.

1.17.2 Recording and justifying design ideas clearly and effectively

REVISED ☐

- Designers must be able to justify their ideas and the design decisions that they have taken, for example giving reasons why certain materials and components have been chosen. This should always be in terms of the needs and wants of the end user and how well each design would meet them.
- This can be achieved through detailed annotations of design ideas, or listing how each one meets the different requirements of the design specification. Consideration should be given to the wider design context and how each idea fits within this.

Now test yourself

TESTED ☐

1 Describe the differences between freehand sketching and formal drawing. (4)
2 Name **two** types of perspective drawings. (2)
3 Explain the differences between first and third angle orthographic projections. (4)
4 Describe the purpose of a systems diagram. (2)
5 Give **two** reasons why CAD is used to produce drawings of products. (2)